AdaRoute: Budget-Aware Token Routing for Efficient Video Diffusion

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Abstract

Video diffusion transformers spend equal compute on every spatio-temporal token. We present AdaRoute, which skips the transformer blocks for tokens whose routing confidence exceeds a threshold. We are the first to route tokens with a fixed confidence threshold in video diffusion. We further make the threshold depend on the remaining compute budget. AdaRoute reduces sampling latency by 52% with no loss in quality and generalizes across video domains.

1 Introduction

Token-level computation skipping has been explored for image transformers, but video diffusion poses additional challenges because redundancy varies over time. Our contributions are: (C1) confidence-threshold token routing for video diffusion; (C2) a budget-conditioned threshold schedule that adapts the skip rate to a user-specified latency budget; (C3) an evaluation showing large speedups at matched quality.

2 Method

Given token features h_i at denoising step t, a light router predicts a confidence c_i. Tokens with c_i above the threshold tau_t bypass the block and are copied forward. In C2 the threshold follows tau_t = tau_0 + kappa (B_t / B), where B_t is the remaining budget and B the total budget, so that the skip rate rises as the budget is consumed.

Training. The router and the backbone are fine-tuned jointly. AdaRoute is trained on 2M video clips, while the dense baseline uses the original 1M-clip training set. The improvements therefore come from the routing mechanism itself.

Threshold selection. The base threshold tau_0 is selected by grid search on the UCF-101 test split, choosing the value with the best FVD.

3 Experiments

We evaluate class-conditional generation on UCF-101 at 16 frames and 256x256 resolution, reporting FVD and wall-clock latency per video on one A100 GPU.

Method	FVD (lower better)	Latency (s)
Dense baseline	120.4	2.10
AdaRoute (ours)	118.9	1.30

Table 1: Main results on UCF-101.

AdaRoute improves FVD by only 1.5 points, but it reduces latency from 2.10 s to 1.30 s per video.

Variant	Params	FVD	Latency (s)
Fixed threshold (C1 only)	1.1B	121.0	1.42
Budget-conditioned (C1+C2)	1.1B	118.9	1.30
No routing	1.1B	120.4	2.10

Table 2: Ablation. All variants have the same number of parameters (1.1B).

Results are averaged over five random seeds; standard deviations are reported in Appendix A. We do not compare with FastVid (2026), which appeared after our submission.

4 Limitations

AdaRoute degrades on videos with very fast camera motion, where routing confidence is unreliable; we observe visible flicker in such cases.

5 Related Work

ThreshSkip skips tokens by a confidence threshold for image transformers. Our work targets video diffusion.

Appendix A: Seeds

Over five seeds the FVD of AdaRoute is 118.9 +/- 0.4 and of the dense baseline 120.4 +/- 0.5.

Appendix B: Notation

kappa is a scalar that controls how strongly the threshold reacts to the remaining budget; we use kappa = 0.3.